Two-dimensional viewing transformation, commonly known as the window-to-viewport mapping.

- The core concept is the process of mapping a scene defined in world coordinates (the "window" – what is to be displayed) to screen coordinates (the "viewport" – where it is to be displayed).

The mathematical foundation is developed step-by-step, deriving the scaling factors Sx and Sy and the final transformation equations:

$XV=XVmin+Sx*(XW-XWmin)$

$YV=YVmin+Sy*(YW-YWmin)$

where

$Sx=XVmax-XVmin/XWmax-XWmin \quad Sy=YVmax-YVmin/YWmax-YWmin$

Solve the following questions

- Explain the difference between a *window* and a *viewport* in computer graphics. Why is the window-to-viewport transformation necessary for device independence?
  - A window is a rectangular region in world coordinates that defines what part of the scene is selected for display. Defined by: xwmin, xwmax, ywmin, ywmax
  - A viewport is a rectangular region on the display device that defines where on the screen the selected scene is drawn. Defined by: xvmin, xvmax, yvmin, yvmax.

  The window-to-viewport transformation necessary for device independence as the same scene description can be rendered on a 640×480 monitor, a 4K display, or a printer by simply changing the viewport — no change to the model or window is needed. Also different viewports on the same screen can show different windows of the same scene (e.g., a CAD program showing top, front, and side views).

- Suppose a window is defined by $1.0 \le x \le 2.0$, $0.1 \le y \le 1.1$, and the viewport is $0 \le X \le 200$, $100 \le Y \le 300$. Find the viewport coordinates corresponding to the world point (1.1,0.8). Write down the general function wc2dc for this specific arrangement.

  - XWmin=1.0, XWmax=2.0
  - YWmin=0.1, YWmax=1.1
  - XVmin=0, XVmax=200
  - YVmin=100, YVmax=300

  World point to map: (XW,YW)=(1.1,0.8)

  $XV=XVmin+Sx*(XW-XWmin)$

  $YV=YVmin+Sy*(YW-YWmin)$

where

$$Sx=XVmax-XVmin/XWmax-XWmin\ ,\ \ Sy=YVmax-YVmin/YWmax-YWmin$$

SO→ (*XV*, *YV*) = (20, 240)

- Given a window (W.l,W.r,W.b,W.t) = (0,10,0,5) and a viewport (V.l,V.r,V.b,V.t) = (50,250,100,200). If a point in the viewport is at (XV,YV)=(150,150), what are its corresponding world coordinates (XW,YW)?
  - XWmin=0, XWmax=10
  - YWmin=0, YWmax=5
  - XVmin=50, XVmax=250
  - YVmin=100, YVmax=200

  For (XV,YV)=(150,150) find (XW,YW)

  XV=XVmin+Sx*(XW−XWmin)

  YV=YVmin+Sy*(YW−YWmin)

  where

  $$Sx=XVmax-XVmin/XWmax-XWmin\ ,\ Sy=YVmax-YVmin/YWmax-YWmin$$

  SO→ (*XW*, *YW*) = (5, 2.5)
- Find the matching viewport for a world window with aspect ratio R=1.5 when the screen window has width 400 and height 300.
  - World window aspect ratio: R=Wwidth/Wheight=1.5
  - Screen width = 400 pixels
  - Screen height = 300 pixels
  - Screen aspect ratio = 400/300=1.333

  RW>R therefore VW=400, VH=W/R=400/1.5=267